

3.jQuery

jQuery is a fast and concise JavaScript Library created by John Resig in 2006 with a nice motto: **Write less, do more.** jQuery simplifies HTML document traversing, event handling, animating, and Ajax interactions for rapid web development.

jQuery Features

jQuery simplifies various tasks of a programmer by writing less code. Here is the list of important core features supported by jQuery –

- **DOM manipulation** – The jQuery made it easy to select DOM elements, negotiate them and modifying their content by using cross-browser open source selector engine called **Sizzle**.
- **Event handling** – The jQuery offers an elegant way to capture a wide variety of events, such as a user clicking on a link, without the need to clutter the HTML code itself with event handlers.
- **AJAX Support** – The jQuery helps you a lot to develop a responsive and featurerich site using AJAX technology.
- **Animations** – The jQuery comes with plenty of built-in animation effects which you can use in your websites.
- **Lightweight** – The jQuery is very lightweight library - about 19KB in size (Minified and gzipped).
- **Cross Browser Support** – The jQuery has cross-browser support, and works well in IE 6.0+, FF 2.0+, Safari 3.0+, Chrome and Opera 9.0+
- **Latest Technology** – The jQuery supports CSS3 selectors and basic XPath syntax.

Basic Syntax

The basic jQuery syntax is:

javascript

Copy code

```
$(selector).action();
```

- **\$**: The jQuery function.
- **selector**: Identifies the element(s) to manipulate (e.g., #id, .class).
- **action()**: The method or event to execute.

Example:

javascript

Copy code

```
$(document).ready(function () {  
    $("#button").click(function () {  
        alert("Button clicked!");  
    });  
});
```

Core Features of jQuery

1. DOM Manipulation

- **Selecting Elements:**

```
javascript
Copy code
$("#id");      // Select by ID
$(".class");   // Select by class
$("tagname");  // Select by tag
```

- **Modifying Content:**

```
javascript
Copy code
$("#element").text("New text");    // Set text content
$("#element").html("<b>HTML</b>"); // Set HTML content
$("#element").val("value");        // Set input value
```

- **Adding/Removing Elements:**

```
javascript
Copy code
$("#list").append("<li>Item</li>"); // Add at the end
$("#list").prepend("<li>Item</li>"); // Add at the start
$("#item").remove();                // Remove element
```

2. Event Handling

- **Common Event Methods:**

```
javascript
Copy code
$("button").click(function () { ... }); // On click
$("input").focus(function () { ... }); // On focus
$("input").blur(function () { ... });   // On blur
```

- **General Event Binding:**

```
javascript
Copy code
$("selector").on("event", childSelector, function() { ... });
```

Example:

```
javascript
Copy code
$(document).on("click", ".dynamic-element", function () {
    alert("Dynamic element clicked!");
});
```

```
});
```

3. CSS Manipulation

- **Adding/Removing Classes:**

```
javascript
Copy code
$("#element").addClass("highlight"); // Add class
$("#element").removeClass("highlight"); // Remove class
```

- **Changing CSS Properties:**

```
javascript
Copy code
$("#element").css("color", "blue"); // Change text color
```

4. AJAX with jQuery

jQuery simplifies AJAX calls to communicate with the server without refreshing the page.

- **GET Request:**

```
javascript
Copy code
$.get("url", function (data) {
    console.log(data);
});
```

- **POST Request:**

```
javascript
Copy code
$.post("url", { key: "value" }, function (data) {
    console.log(data);
});
```

- **Full AJAX Request:**

```
javascript
Copy code
$.ajax({
    url: "url",
    method: "POST",
    data: { key: "value" },
    success: function (response) {
        console.log(response);
    },
    error: function (error) {
```

```
        console.error(error);
    }
});
```

5. Effects and Animations

- **Show/Hide Elements:**

```
javascript
Copy code
$("#element").show(); // Show element
$("#element").hide(); // Hide element
```

- **Fading Effects:**

```
javascript
Copy code
$("#element").fadeIn(); // Fade in
$("#element").fadeOut(); // Fade out
```

- **Sliding Effects:**

```
javascript
Copy code
$("#element").slideDown(); // Slide down
$("#element").slideUp(); // Slide up
```

- **Custom Animations:**

```
javascript
Copy code
$("#element").animate({ opacity: 0.5, height: "50px" }, 1000);
```

6. Traversing the DOM

- **Find Child Elements:**

```
javascript
Copy code
$("#parent").find(".child");
```

- **Move Up/Down:**

```
javascript
Copy code
$(".child").parent(); // Move up to parent
$(".child").siblings(); // Get sibling elements
```

Example: Form Validation with jQuery

html

Copy code

```
<form id="myForm">
  <input type="text" id="name" placeholder="Name">
  <input type="email" id="email" placeholder="Email">
  <button type="submit">Submit</button>
</form>

<script>
  $(document).ready(function () {
    $("#myForm").on("submit", function (e) {
      e.preventDefault(); // Prevent form submission
      const name = $("#name").val();
      const email = $("#email").val();

      if (!name || !email) {
        alert("All fields are required!");
      } else {
        alert("Form submitted successfully!");
      }
    });
  });
</script>
```